

Caden Cheng

La Jolla, CA | 408-592-6669 | cfc005@ucsd.edu | linkedin.com/in/cadenfcheng | github.com/cadencheng888

EDUCATION

University of California, San Diego

La Jolla, CA

B.S. in Computer Science and Mathematics, Data Science Minor

Expected Grad 2028

GPA: 3.53 | **Relevant Coursework:** Data Structures, Discrete Math, Systems Programming, Algorithms

EXPERIENCE

Software Developer → Incoming VP of Technology & Engineering Manager Oct. 2025 – Present

Computer Science Engineering Society (CSES) at UCSD

La Jolla, CA

- Onboarded **100 users** across **10 UCSD labs** by shipping the auth, role-selection, and profile-setup in **Next.js**
- Secured API access for **4 user roles** by building an **RBAC** layer with **10 scoped permissions** in **TypeScript**
- Hardened auth reliability by writing **11 Jest unit tests** covering getSession() permission and status checks
- Restricted **10 labs'** data to verified @ucsd.edu users by integrating **Google OAuth** with **NextAuth** and **JWT**
- Modeled user and role data by designing a **Mongoose** User schema with email validation and indexed lab info

Software Developer

Mar. 2026 – Present

Triton Web Developers (TWD) at UCSD

La Jolla, CA

- Empowered **100+ club members** as measured by **40+ content updates** without a developer using **TinaCMS**
- Modeled site content as measured by **9 typed CMS collections** (hero, about, FAQ, board, gallery, socials)
- Served **100+ members** as measured by a self-serve FAQ accordion built in **React** with expand/collapse state
- Showcased membership value as measured by a responsive section with **3 CMS-driven benefit tiers**

PROJECTS

inVISION | *Python, OpenCV, MediaPipe, NumPy, Pygame*

Apr. 2026

- Won **Best Interactive AI** at ACM Diamond Hacks 2026, largest hackathon in San Diego (**500+ participants**)
- Developed a real-time 3D model generator and interactive viewer powered by custom computer vision pipelines
- Automated the conversion of 2D images into 3D geometries by integrating the **TripoSR** framework
- Built an optimized 3D rendering pipeline using **NumPy** for transformations, culling, and lighting internally
- Implemented a computer vision control scheme using **MediaPipe** for native pinch-to-zoom and rotation tracking

Silent Speech | *PyTorch, OpenCV, TensorFlow, Python*

Jan. 2026

- Won **#1 Best Overall Hack** at San D Hacks 2026 out of 35+ competing undergraduate teams
- Developed a real-time lip-reading AI using a **BiGRU**, achieving **85% accuracy** on live webcam feeds
- Engineered a pipeline using **OpenCV** for facial landmark extraction, normalizing features for scale invariance
- Optimized model training using stratified sampling, mixup, and noise injection data augmentation

remark. | *Python, TypeScript, React, Claude API, Deepgram, FastAPI*

Jun. 2026

- Shipped a hands-free AI voice agent in **24h** at **Berkeley AI Hackathon 2026**, owning audio + HUD
- Routed **95%** of spoken intents correctly across **10+ task types** by wiring a **3-tier** agentic router
- Cut speech-to-action latency to **2.5s** by building a real-time **React** HUD streaming captions and reasoning
- Protected user privacy by coding an on-device **OpenCV** face-gate that limits passive listening to active talk

ACM Projects Spotify Song Recommender | *Pandas, PyTorch, NumPy, Matplotlib*

Jan. 2026 – Apr. 2026

- Developed a collaborative filtering recommendation system in **PyTorch** to predict user audio preferences
- Processed and structured a massive dataset of **7.7M+ interactions** spanning 930K+ users and 30K+ songs
- Engineered a custom PyTorch DataLoader pipeline with negative sampling to efficiently generate balanced batches
- Optimized weights using Adam and MSE loss, evaluating generalization via Precision, Recall, and ROC AUC

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, C/C++, HTML/CSS, SQL

Frameworks: React, Next.js, Node.js, Tailwind CSS, PyTorch, TensorFlow

Developer Tools: Git/GitHub, MongoDB Atlas, npm, Jest

Libraries: Pandas, NumPy, Matplotlib, Torchvision, OpenCV, MediaPipe, Mongoose